

STUDENT PERFORMANCE ANALYTICS DASHBOARD USER

GUIDE

Mega Central High School

Version: 1.0

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1. Introduction

The **Student Performance Analytics Dashboard** is an interactive Business Intelligence solution developed to help Mega Central High School monitor academic performance, identify students requiring intervention, evaluate curriculum effectiveness, and support evidence-based educational decision-making.

The solution integrates assessment data, student information, question-level analytics, and performance metrics into four interactive Power BI dashboards, enabling stakeholders to monitor school performance at both strategic and operational levels.

2. Project Objective

The objectives of the dashboard are to:

- Monitor overall school academic performance

- Track student performance across assessments
- Identify academically at-risk students
- Compare subject-wise performance
- Identify weak chapters and learning gaps
- Evaluate assessment difficulty
- Support teachers with targeted interventions
- Enable data-driven academic decision-making

3. Intended Users

Principal

Uses the dashboard to:

- Monitor school-wide academic performance
- Track pass percentage
- Review overall student risk profile
- Monitor monthly performance trends
- Support strategic academic planning

Academic Coordinator

Uses the dashboard to:

- Compare subject performance
- Monitor chapter-level accuracy
- Identify curriculum gaps
- Review assessment quality
- Plan remedial programs

Teachers

Uses the dashboard to:

- Monitor individual student performance
- Review assessment history
- Identify weak subjects and chapters
- Track student progress over time
- Plan personalized interventions

4. Dashboard Overview

The solution contains **four interactive dashboards**.

Dashboard	Purpose
Executive Performance Overview	School-level academic KPIs and overall performance trends
Individual Student Performance Analysis	Detailed academic profile of an individual student

Risk Analysis Dashboard	School-wide risk monitoring and identification of high-risk students
Subject & Chapter Performance Analysis	Evaluation of curriculum effectiveness and chapter-level learning outcomes

5. Dashboard Navigation

Users can switch between dashboards using the report tabs at the bottom of the Power BI report.

Navigation sequence:

Executive Performance Overview



Individual Student Performance Analysis



Risk Analysis Dashboard



Subject & Chapter Performance Analysis

6. Executive Performance Overview

Purpose: Provides a high-level summary of school performance for leadership and academic management.

KPI Cards

The dashboard displays:

- Total Students
- Tests Conducted
- Average Score
- Pass Percentage
- High Risk Students
- Top Performers

These KPIs provide an instant overview of the school's academic health.

Subject Performance

Displays average percentage achieved in each subject.

Subjects include:

- English
- Mathematics
- Science
- Social

Business Use

- Compare subject performance
- Identify strongest and weakest subjects
- Support curriculum planning

Student Performance Distribution

Shows distribution of students across performance bands:

- Top Performer
- High Performer
- Average
- Needs Support
- At Risk

Business Use

Helps management understand the overall academic performance distribution across the school.

Overall Student Risk Profile

Displays the proportion of students classified as:

- Low Risk
- Medium Risk
- High Risk

Business Use

Supports prioritization of academic interventions.

Monthly Academic Performance Trend

Displays the monthly average academic performance from June to January.

A target performance line (70%) is included for comparison.

Business Use

- Monitor school performance over time
- Identify improvement or decline
- Compare actual performance against target

7. Individual Student Performance Analysis

Purpose: Provides a comprehensive academic profile of a selected student.

Student Selection

A dropdown slicer allows users to select any student.

Selecting a student updates all visuals automatically.

Student Information Cards

Displays:

- Student Name
- Grade
- Risk Level
- Performance Band
- Average Score

These cards provide a quick summary of the student's academic status.

Subject Performance Analysis

Displays the student's average score across:

- English
- Mathematics
- Science
- Social

Business Use

Identifies the student's strongest and weakest subjects.

Chapter-wise Learning Gaps

Displays the lowest-performing chapters for the selected student.

The visualization highlights the chapters requiring immediate attention.

Business Use

Supports personalized remedial teaching.

Student Performance Trend

Shows the student's average assessment performance over time.

A target score of **70%** is displayed for comparison.

Business Use

Monitors academic improvement across assessments.

Assessment History

Displays detailed assessment records including:

- Test ID
- Date
- Subject
- Test Type
- Score Percentage
- Grade
- Pass / Fail

Business Use

Provides teachers with complete assessment history for academic review.

8. Risk Analysis Dashboard

Purpose: Provides school-wide visibility into academic risk and identifies students requiring intervention.

KPI Cards

Displays:

- Total Students
- High Risk Students
- Medium Risk Students
- Low Risk Students
- Average Score
- Students Needing Academic Support

These indicators summarize the school's overall risk profile.

Overall Student Risk Profile

Displays the proportion of:

- Low Risk
- Medium Risk
- High Risk

Business Use

Measures overall academic risk across the school.

Attendance Category Distribution

Shows the number of students categorized as:

- Excellent
- Good
- Average
- Poor

Business Use

Helps evaluate attendance patterns alongside academic performance.

Subject-wise Risk Assessment

Compares pass and fail percentages across all subjects.

Subjects include:

- English
- Mathematics
- Science

- Social

Business Use

Identifies subjects contributing most to academic risk.

High Risk Students List

Displays:

- Student Name
- Grade
- Risk Level
- Average Percentage
- Attendance Category

Business Use

Provides an actionable intervention list for teachers and administrators.

9. Subject & Chapter Performance Analysis

Purpose: Evaluates subject performance, chapter effectiveness, and assessment quality.

KPI Cards

Displays:

- Total Subjects
- Total Chapters
- Average Accuracy
- Chapters Below Target

These KPIs summarize curriculum performance.

Subject-wise Performance

Displays average accuracy across:

- English
- Mathematics
- Science
- Social

Business Use

Supports comparison of subject-level learning outcomes.

Question Difficulty Distribution

Displays the number of questions categorized as:

- Easy
- Medium
- Hard

The chart is further segmented by subject.

Business Use

Evaluates whether assessment difficulty is appropriately balanced.

Chapters Requiring Attention

Displays the lowest-performing chapters based on average accuracy.

Business Use

Identifies chapters requiring remedial instruction or curriculum review.

Chapter Accuracy Distribution

Displays average accuracy for individual chapters.

Business Use

Helps compare learning outcomes across the curriculum.

Chapter Performance Table

Displays:

- Chapter Name
- Subject
- Average Accuracy Rate

Business Use

Provides detailed chapter-level performance information for academic planning.

10. Dashboard Interactivity

The dashboards provide interactive filtering and cross-highlighting.

Supported interactions include:

Student Selection

Updates:

- KPI Cards
- Charts
- Tables

Cross Filtering

Selecting any visual automatically filters related visuals on the page.

Tooltips

Hovering over visuals displays additional contextual information.

Drill-through (if enabled)

Allows detailed analysis from summary-level visuals.

11. Key Performance Indicators

KPI	Description
Average Score	Average percentage across all students
Pass Percentage	Percentage of students achieving the minimum passing score
High Risk Students	Students requiring immediate academic intervention
Performance Band	Overall academic performance classification
Average Accuracy	Average chapter-level correctness
Chapters Below Target	Chapters with accuracy below the defined threshold

12. Business Insights

The dashboards help answer important academic questions, including:

- Which students require immediate intervention?
- Which subjects are underperforming?
- Which chapters have the lowest learning outcomes?
- How is academic performance changing over time?
- Which assessments are most difficult?
- What proportion of students fall into each risk category?
- Which students require additional academic support?

13. Data Sources

The solution uses synthetic educational datasets created for portfolio demonstration purposes.

The following datasets are included:

- Students
- Tests
- Student Responses
- Question Bank
- Assessment Scores
- Chapter Analytics
- Student Risk Classification
- Performance Band Classification

The datasets are processed through a Python ETL pipeline, stored in Microsoft SQL Server, and visualized using Power BI.

14. Business Rules

Pass Criteria

A student is considered **Pass** if: **Percentage $\geq$ 35%**

Otherwise: **Fail**

Performance Bands

Students are classified into:

- Top Performer
- High Performer
- Average
- Needs Support
- At Risk

Risk Levels

Students are categorized as:

- Low Risk
- Medium Risk
- High Risk

Target Score

The expected academic performance target is: **70%**

15. Assumptions & Limitations

- This project uses synthetic educational data generated solely for demonstration purposes.
- Student names, assessment records, attendance categories, and scores are fictional.
- Attendance values are simulated and do not represent actual school records.
- The dashboards are intended to demonstrate Business Intelligence and Business Analysis capabilities rather than operational reporting.

16. Future Enhancements

Potential enhancements include:

- Parent Portal
- Teacher Performance Dashboard
- Attendance Analytics Dashboard
- Predictive Student Performance Modeling
- AI-based Risk Prediction
- Personalized Learning Recommendations
- Automated Intervention Notifications
- Learning Management System (LMS) Integration
- Real-time SQL Database Connectivity
- Mobile Power BI Dashboard

- District and Multi-school Performance Comparison